CS5004, Spring 2026

Lab 7: Java Collections Framework. Interfaces `Iterable` and `Comparable`.

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1. Summary

In today's lab, we will continue our conversation about Java Collections Framework. More specifically, we will focus on:

- Interfaces `Iterable` and `Iterator`
- Interfaces `Comparable` and `Comparator`
- Maps and sets
- Graphs

Note 1: Labs are intended to help you get started, and give you some practice while the course staff is present and able to provide assistance. You are not required to finish all the questions during the lab, but you are expected to push your lab work to a designated repo on the Khoury GitHub.

Problem 1: Data Collections

Implement a method that consumes a `List<Integers>`, and returns a frequency count of the numbers in the input list as a `Map`. A frequency count contains each number *n* in the list, and the **relative frequency** of time that *n* appears in the list. Relative frequency is computed as the number of occurrences of *n* divided by the total number of occurrences in the input list.

For example, given the list of numbers:

(1, 1, 1, 2, 2, 3, 3, 4, 4, 4)

the frequency count should contain the following information:

1 → 0.3
2 → 0.2
3 → 0.2
4 → 0.3

Problem 2: Interface `Comparable`

Consider class `Swimmer` given below. Notice that the given class implements interface `Comparable`. Implement method `compareTo(Swimmer otherSwimmer)`, that compares two `Swimmer` objects based upon their average freestyle swimming times, computed as a sum of all their freestyle swimming times, and divided by the number of their recorded freestyle times.

```java
import java.util.List;

/*Class Swimmer contains information about a swimmer, and the list of their swimming times, when
swimming 50m. Four swimming styles are considered: butterfly, backstroke, breaststroke, and freestyle.
 */
public class Swimmer implements Comparable<Swimmer> {

  private String name;
  private List<Double> butterfly50mTimes;
  private List<Double> backstroke50mTimes;
  private List<Double> breaststroke50mTimes;
  private List<Double> freestyle50mTimes;

  public Swimmer(String name, List<Double> butterfly50mTimes, List<Double> backstroke50mTimes,
                              List<Double> breaststroke50mTimes, List<Double> freestyle50mTimes) {
    this.name = name;
    this.butterfly50mTimes = butterfly50mTimes;
    this.backstroke50mTimes = backstroke50mTimes;
    this.breaststroke50mTimes = breaststroke50mTimes;
    this.freestyle50mTimes = freestyle50mTimes;
  }

  public String getName() {
    return name;
  }

  public List<Double> getButterfly50mTimes() {
    return butterfly50mTimes;
  }

  public List<Double> getBackstroke50mTimes() {
    return backstroke50mTimes;
  }

  public List<Double> getBreaststroke50mTimes() {
    return breaststroke50mTimes;
  }

  public List<Double> getFreestyle50mTimes() {
    return freestyle50mTimes;
  }

  @Override
  //YOUR CODE HERE
  public int compareTo(Swimmer otherSwimmer) {
  }
}
```

Problem 3: Interface `Iterator`

Given a list of swimmers,

```
List<Swimmer> runners = new ArrayList<>();
```

implement an iterator `SwimmersIterator` that iterates over the given list of swimmers, and returns those swimmers who have at least five butterfly swim times, and who have achieved at least one 50m Olympic freestyle qualifying time. The current 50m freestyle Olympic qualifying time is 26.17s.

Note: You do not have to implement method `remove()`.